SFT V11.7 - Overlapping Lag Fields and Fluid Stress

Seed note v6: constitutive derivation plus 1D/2D robustness closure

Status. This document is a theoretical and numerical seed note. It does not claim to solve Navier-Stokes or establish new experimental physics. It records an SFT-motivated effective-fluid stress closure and the toy-model robustness checks completed through v6.

Core v6 upgrade. The previous numerical closure is now paired with a constitutive derivation: coarse-grained lag overlap decomposes, by symmetry, into an isotropic scalar compressive field and a traceless tensor shear-memory field. The coefficients remain phenomenological, but their role is now tied to a cost functional and relaxation dynamics.

Prepared as part of the SFT V11 working sequence.

SFT V11.7 Fluid Stress Seed v6 - final effective-theory update

1. Executive Summary

V11.7 began with a simple question: if SFT lag fields overlap inside matter, do they behave like an effective stress contribution in fluids? The toy program now supports a sharper answer. In 1D, lag memory behaves as a finite-memory stress reservoir organized by a nondimensional lag-memory number. In 2D, the stress separates into scalar compressive and tensor shear-memory channels.

$$\Lambda = \frac{\varepsilon\,\alpha\,\tau}{D}$$

$$\boldsymbol{\sigma}_{\mathrm{SFT}} = -\varepsilon_{\chi}\chi\mathbf{I} + \varepsilon_{Q}\mathbf{Q}_{\chi}$$

Here, chi is the scalar compressive lag-pressure channel, and Q_chi is a symmetric traceless tensor lag field that couples to shear strain and vorticity through div Q_chi.

Validation item	v6 status
1D Lambda collapse	Passed
1D resolution robustness	Passed
1D initial-condition robustness	Passed
1D ordinary-viscosity separation	Passed
2D scalar/tensor architecture	Passed
2D resolution robustness	Passed
2D initial-condition robustness	Passed
2D ordinary-viscosity separation	Passed
Constitutive derivation of closure	Added in v6
Coefficient derivation from microscopic lag geometry	Open
3D extension and external benchmark	Future

The numerical phase now supports the effective-fluid seed claim: SFT lag stress can be modeled as a memory-bearing stress correction with distinct scalar and tensor sectors. The coefficient derivation from microscopic SFT lag structure remains the central theoretical gap, not a blocker for the present effective-theory note.

2. Effective-Stress Formulation

Known fluid physics remains primary. The SFT contribution is added as an effective stress correction, not as a replacement for pressure, viscosity, equation of state, thermal physics, molecular forces, or electromagnetic interactions.

$$\boldsymbol{\sigma}_{\mathrm{total}} = \boldsymbol{\sigma}_{\mathrm{known}} + \boldsymbol{\sigma}_{\mathrm{SFT}}$$

$$\boldsymbol{\sigma}_{\mathrm{SFT}} = -\varepsilon_{\chi}\chi\mathbf{I} + \varepsilon_{Q}\mathbf{Q}_{\chi}$$

The scalar term acts as a lag-pressure correction. In multiple dimensions its force is a gradient force, so it does not directly create vorticity. The tensor term is symmetric and traceless; it is the channel that can couple to shear and vorticity through the divergence of a stress tensor.

Minimal evolution equations

$$\partial_t\chi = D_{\chi}\nabla^2\chi + \alpha_{\chi}(\rho - \langle\rho\rangle) - \chi/\tau_{\chi}$$

$$\partial_t\mathbf{Q}_{\chi} = D_{Q}\nabla^2\mathbf{Q}_{\chi} + \alpha_{Q}\mathbf{S} - \mathbf{Q}_{\chi}/\tau_{Q}$$

$$\mathbf{f}_{\mathrm{SFT}} = -\varepsilon_{\chi}\nabla\chi + \varepsilon_{Q}\nabla\cdot\mathbf{Q}_{\chi}$$

The tensor source S is the traceless strain tensor. This is the minimal extension required once compression and shear no longer share the same axis.

3. Constitutive Derivation of the Scalar/Tensor Closure

The v6 theoretical addition is the constitutive derivation of the closure. The claim is not that the coefficients are predicted from first principles. The claim is more modest and more appropriate for an effective theory: once overlapping lag fields are coarse-grained into a local stress response, the lowest-order near-equilibrium closure decomposes into scalar compression and traceless tensor shear by symmetry.

3.1 Coarse-grained lag overlap

Let M_chi denote the coarse-grained lag-overlap tensor inside a fluid cell. Its lowest-order decomposition is the standard trace/traceless split:

$$\mathbf{M}_\chi = \frac{1}{d}\,(\mathrm{Tr}\,\mathbf{M}_\chi)\mathbf{I} + \left(\mathbf{M}_\chi - \frac{1}{d}(\mathrm{Tr}\,\mathbf{M}_\chi)\mathbf{I}\right)$$

$$\chi \propto \mathrm{Tr}\,\mathbf{M}_\chi, \qquad \mathbf{Q}_\chi \propto \mathbf{M}_\chi - \frac{1}{d}(\mathrm{Tr}\,\mathbf{M}_\chi)\mathbf{I}$$

In words: compression changes the amount of lag overlap; shear changes the shape of lag overlap. In 2D the traceless tensor can be represented as

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Q_chi = [[ Q_xx, Q_xy ], [ Q_xy, -Q_xx ]]
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3.2 Lag cost functional

The simplest quadratic cost functional contains a local storage cost, a gradient cost, and source couplings to density, compression, and shear.

$$\mathcal{F}_\chi = \int \left[\frac{a_\chi}{2}\chi^2 + \frac{\kappa_\chi}{2}|\nabla\chi|^2 - g_\rho\chi\,\delta\rho - g_\theta\chi\,\theta\right] d^d x$$

$$\mathcal{F}_Q = \int \left[\frac{a_Q}{2}\mathbf{Q}_\chi : \mathbf{Q}_\chi + \frac{\kappa_Q}{2}|\nabla\mathbf{Q}_\chi|^2 - g_Q\mathbf{Q}_\chi : \mathbf{S}\right] d^d x$$

Here delta rho = rho - , theta = div u, and S is the traceless strain tensor:

$$\mathbf{S} = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^T) - \frac{1}{d}(\nabla\cdot\mathbf{u})\mathbf{I}$$

This makes the source structure explicit: density and compression source the scalar channel, while shear sources the tensor channel.

4. Relaxation Dynamics and Coefficient Meaning

Assume the lag fields relax downhill in the cost functional. For the scalar channel:

$$\partial_t \chi = -\Gamma_\chi \frac{\delta \mathcal{F}}{\delta \chi}$$

This gives the linear response equation

$$\partial_t \chi = D_\chi \nabla^2 \chi + \alpha_\rho \delta\rho + \alpha_\theta \theta - \chi/\tau_\chi$$

with

$$D_\chi = \Gamma_\chi \kappa_\chi, \qquad \alpha_\rho = \Gamma_\chi g_\rho, \qquad \alpha_\theta = \Gamma_\chi g_\theta, \qquad \tau_\chi = (\Gamma_\chi a_\chi)^{-1}$$

Likewise, for the tensor channel:

$$\partial_t \mathbf{Q}_\chi = D_Q \nabla^2 \mathbf{Q}_\chi + \alpha_Q \mathbf{S} - \mathbf{Q}_\chi/\tau_Q$$

$$D_Q = \Gamma_Q \kappa_Q, \qquad \alpha_Q = \Gamma_Q g_Q, \qquad \tau_Q = (\Gamma_Q a_Q)^{-1}$$

The simulation coefficients are therefore no longer merely arbitrary knobs. They map onto physical categories: spatial smoothing of lag overlap, source coupling strength, memory/relaxation time, and conversion from lag field amplitude into mechanical stress.

Coefficient	Effective-theory meaning
D_chi, D_Q	spatial smoothing/correlation transport of lag overlap
alpha_rho, alpha_theta, alpha_Q	source strength from density, compression, or shear
tau_chi, tau_Q	lag-memory relaxation/decorrelation time
epsilon_chi, epsilon_Q	conversion from lag field to mechanical stress/force

4.1 Stress force

The fluid momentum equation receives the divergence of the SFT stress:

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \nabla \cdot \boldsymbol{\tau}_{\text{visc}} + \nabla \cdot \boldsymbol{\sigma}_{\text{SFT}}$$

$$\mathbf{f}_{\text{SFT}} = -\varepsilon_\chi \nabla \chi + \varepsilon_Q \nabla \cdot \mathbf{Q}_\chi$$

5. Fourier Response and Channel Separation

The scalar and tensor channels behave differently because their forces have different vector structure. For a scalar lag field, the Fourier response is

$$\chi(\mathbf{k},\omega) = \frac{\alpha_\rho \delta\rho(\mathbf{k},\omega) + \alpha_\theta \theta(\mathbf{k},\omega)}{\tau_\chi^{-1} + D_\chi k^2 - i\omega}$$

$$\mathbf{f}_\chi(\mathbf{k},\omega) = -i\varepsilon_\chi \mathbf{k}\,\chi(\mathbf{k},\omega)$$

The scalar force points along k. It is longitudinal. Therefore it directly modifies compression and density but has zero curl in the absence of additional baroclinic or thermal structure.

$$\nabla \times (-\nabla\chi) = 0$$

For a tensor lag field:

$$\mathbf{Q}_\chi(\mathbf{k},\omega) = \frac{\alpha_Q \mathbf{S}(\mathbf{k},\omega)}{\tau_Q^{-1} + D_Q k^2 - i\omega}$$

A transverse shear mode u = (0,u_y) exp(ikx) has S_xy = iku_y/2, so the y-component of the tensor force becomes

$$f_{Q,y} = -\frac{\varepsilon_Q \alpha_Q k^2}{2(\tau_Q^{-1} + D_Q k^2 - i\omega)} u_y$$

In the slow-response limit, this is negative feedback on the transverse shear mode:

$$f_{Q,y} \approx -\frac{\varepsilon_Q \alpha_Q k^2}{2(\tau_Q^{-1} + D_Q k^2)} u_y$$

This explains the 2D hero-run signature: increasing tensor lag-memory strength suppresses vorticity RMS, enstrophy, strain, and vorticity high-k power.

6. Control Parameters

In the nondimensional toy normalization, the scalar and tensor lag-memory numbers are

$$\Lambda_\chi = \frac{\varepsilon_\chi \alpha_\chi \tau_\chi}{D_\chi}, \qquad \Lambda_Q = \frac{\varepsilon_Q \alpha_Q \tau_Q}{D_Q}$$

Their meaning is the ratio of memory-bearing lag feedback to spatial lag smoothing. Low Lambda means the lag response relaxes or smooths away before it can alter the flow. Intermediate Lambda gives viscoelastic response. High Lambda gives strong feedback.

Dimensional guardrail. In physical units the exact dimensionless group must include the characteristic length, velocity, density, and timescale used in nondimensionalization. Therefore the toy result should be read as an organizing principle, not yet as a final dimensional law.

Interpretation: Lambda ~ lag-memory feedback / spatial lag relaxation.

The rest of the document summarizes the numerical validation that motivated and then supported this effective-theory closure.

7. 1D Robustness Closure

The v5/v6 1D validation tested 270 nonbaseline runs and 27 baselines with N = 256, 512, 1024; nu = 0.0025, 0.004, 0.008; Lambda = 0.25, 0.75, 1.15, 1.70, 2.40; and three initial-condition families: smooth, gaussian, and random_smooth. All instability flags were false.

Lambda	n	rho std ratio	max |u_x| ratio	high-k ratio	unstable flags
0.25	54	0.986	1.043	1.39	0
0.75	54	0.961	1.131	2.74	0
1.15	54	0.942	1.204	4.45	0
1.70	54	0.920	1.309	7.83	0
2.40	54	0.894	1.445	14.48	0

The 1D signature survives the viscosity sweep: increasing Lambda reduces density contrast while increasing gradient ratio and high-k spectral power. Ordinary viscosity moves the baseline, but it does not remove the Lambda-organized response.

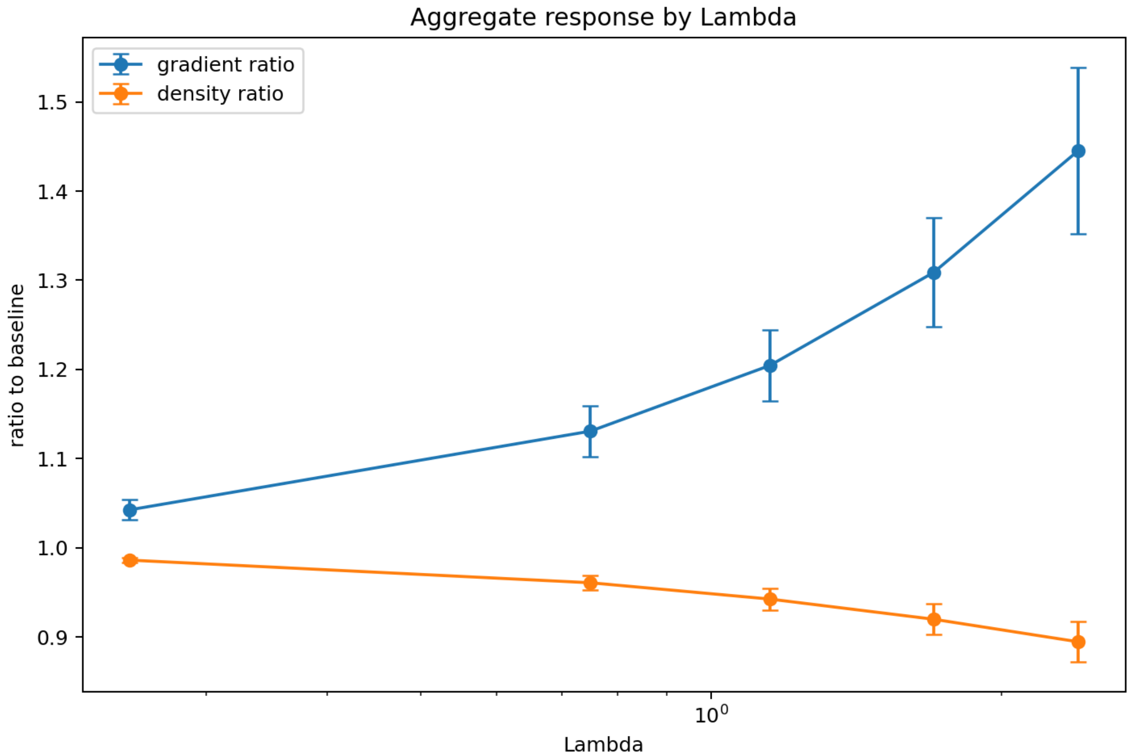

Figure 1. 1D viscosity-separation aggregate response: density contrast falls and gradient ratio rises with Lambda.

8. 1D High-k Feedback and Regime Counts

The inherited label "amplifying_or_unstable" should be read carefully. In the final 1D run it means high-k amplifying feedback, not numerical blow-up, because the instability count was zero.

Regime label	Count
elastic_memory	118
high-k amplifying feedback	73
near_baseline	42
shear_enhancing	37

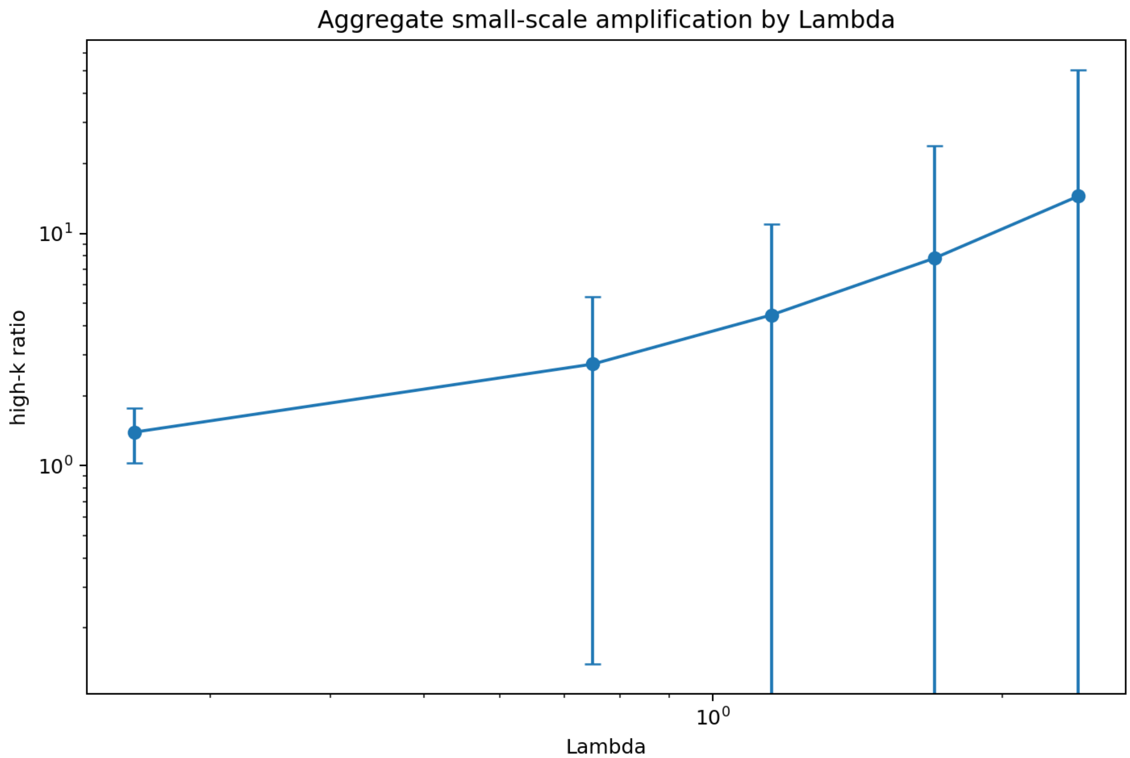

Figure 2. 1D small-scale/high-k response rises with Lambda. Error bars are large because the high-k channel is initial-condition sensitive.

9. 2D Hero Run: Scalar/Tensor Split

The 2D hero run tested 720 nonbaseline runs and 36 baselines with N = 128, 256, 512; nu = 0.0015, 0.0025, 0.005; and four initial-condition families: mixed, vortex_pair, shear_layer, and random_smooth. All instability flags were false.

This run closes the 2D ordinary-viscosity separation check. The scalar/tensor split survives changes in resolution, initial conditions, and ordinary viscosity.

2D regime label	Count
near_baseline	234
shear_memory_damping	225
combined_smoothing	207
compressive_smoothing	54

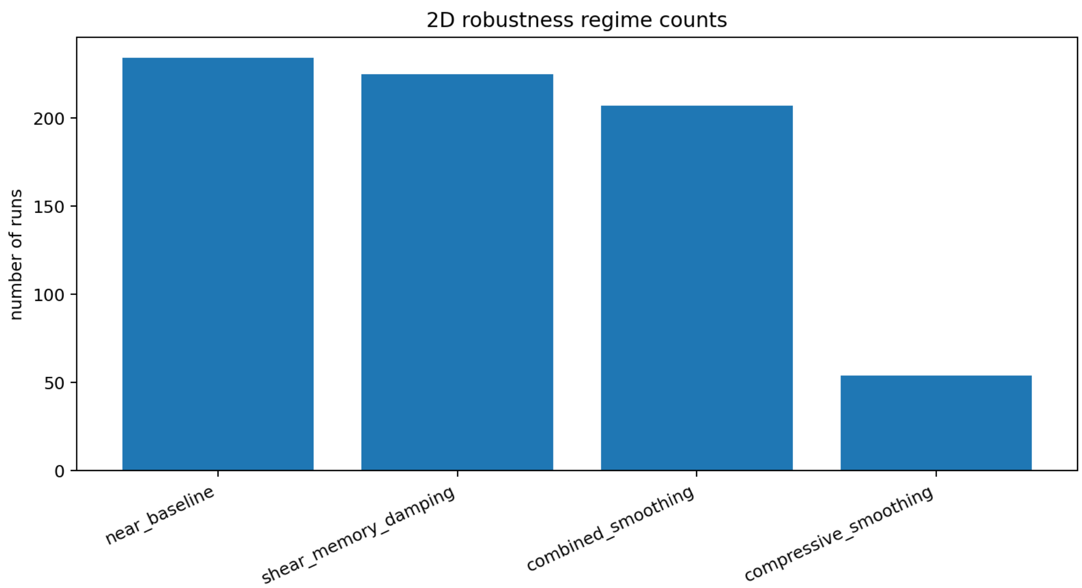

Figure 3. 2D hero run regime counts. Categories track scalar compression, tensor shear damping, combined smoothing, and near-baseline behavior.

10. 2D Scalar Channel: Compressive Lag Pressure

Scalar-only lag stress behaves as a compressive lag-pressure channel. Increasing Lambda_chi reduces density contrast, increases divergence response, and leaves vorticity essentially unchanged. This is the correct vector-calculus behavior for a gradient force.

Lambda_chi	rho std	div RMS	vort RMS	vort high-k
0.25	0.990	1.007	1.000	0.997
0.75	0.970	1.022	1.001	0.995
1.15	0.955	1.035	1.001	1.006
1.70	0.936	1.052	1.002	1.017

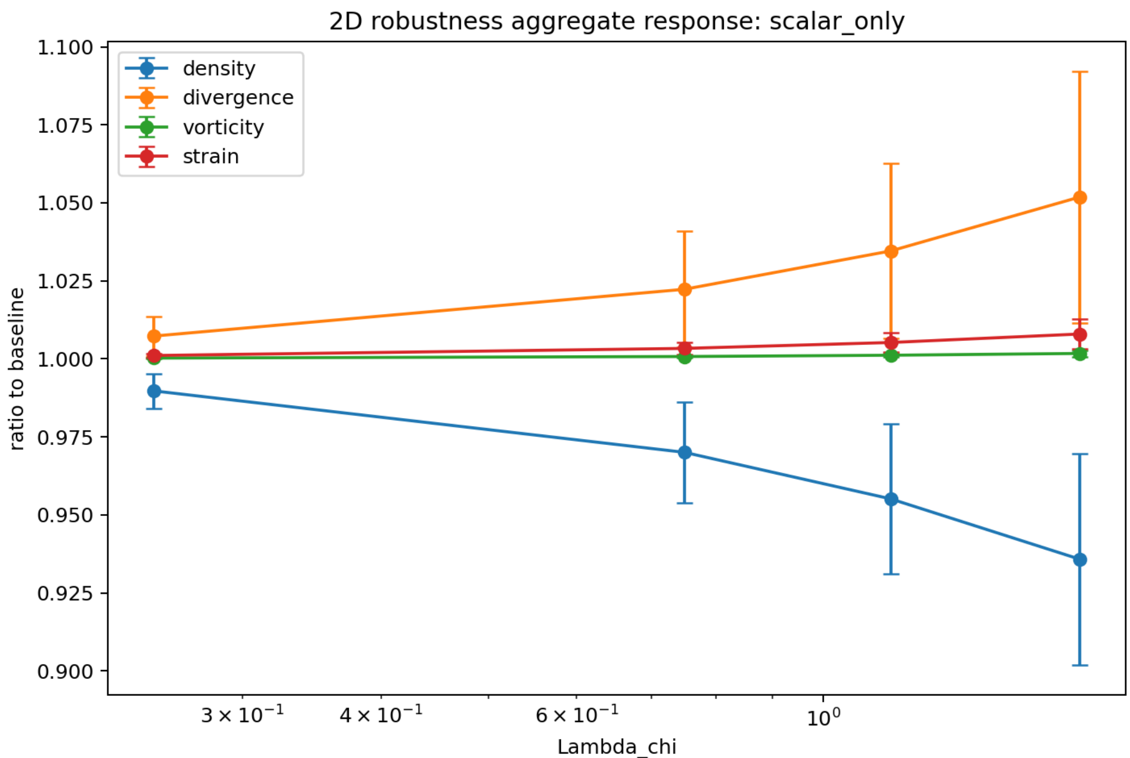

Figure 4. Scalar-only channel: density falls, divergence rises, vorticity remains approximately unchanged.

11. 2D Tensor Channel: Shear-Memory Damping

Tensor-only lag stress is the shear/vorticity channel. Increasing Lambda_Q reduces density contrast, divergence, vorticity RMS, enstrophy, strain, and vorticity high-k power. The result is organized, monotonic, and survives the hero run.

Lambda_Q	rho std	div RMS	vort RMS	enstrophy	vort high-k
0.25	0.986	0.981	0.966	0.933	0.832
0.75	0.960	0.945	0.909	0.830	0.624
1.15	0.940	0.919	0.872	0.766	0.521
1.70	0.915	0.885	0.830	0.697	0.421

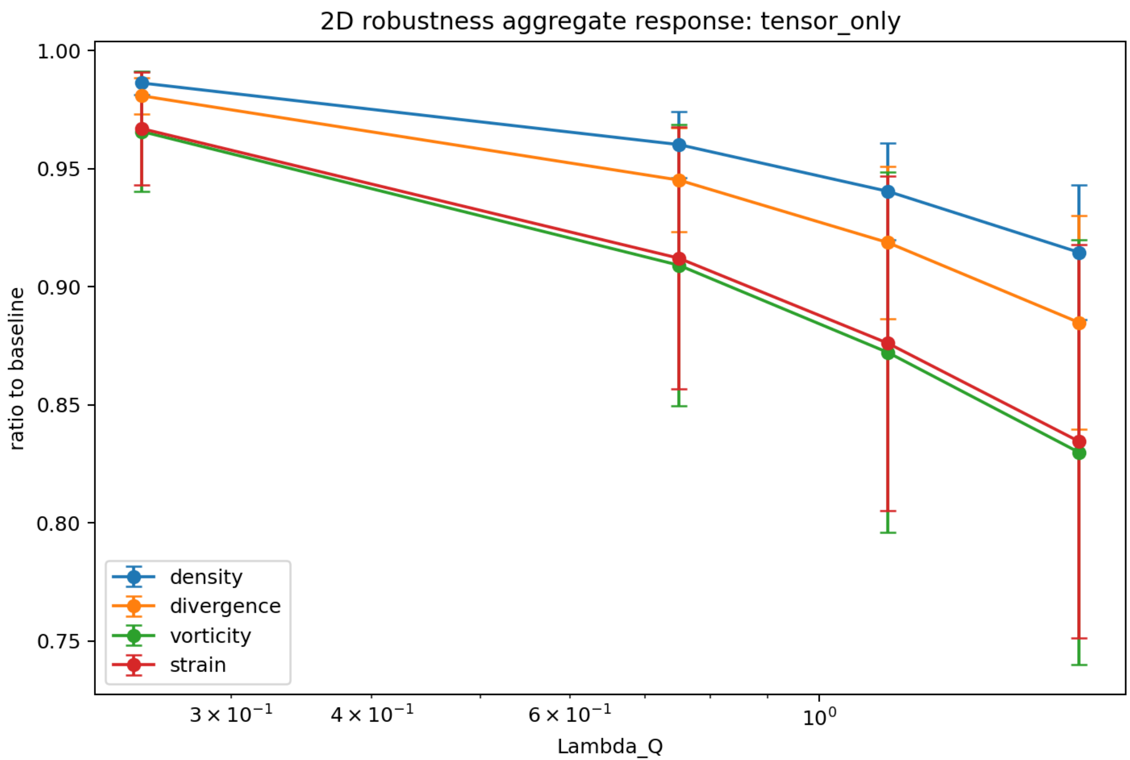

Figure 5. Tensor-only channel: shear, vorticity, divergence, and density ratios fall with Lambda_Q.

12. 2D Combined Channel: Full Stress Architecture

The combined mode sets a fixed scalar channel and varies the tensor channel. It preserves both effects: scalar compression smoothing and tensor shear/vorticity damping.

Lambda_Q	rho std	div RMS	vort RMS	enstrophy	vort high-k
0.25	0.957	1.003	0.967	0.935	0.832
0.75	0.931	0.968	0.910	0.831	0.631
1.15	0.912	0.941	0.873	0.768	0.530
1.70	0.887	0.907	0.831	0.698	0.433

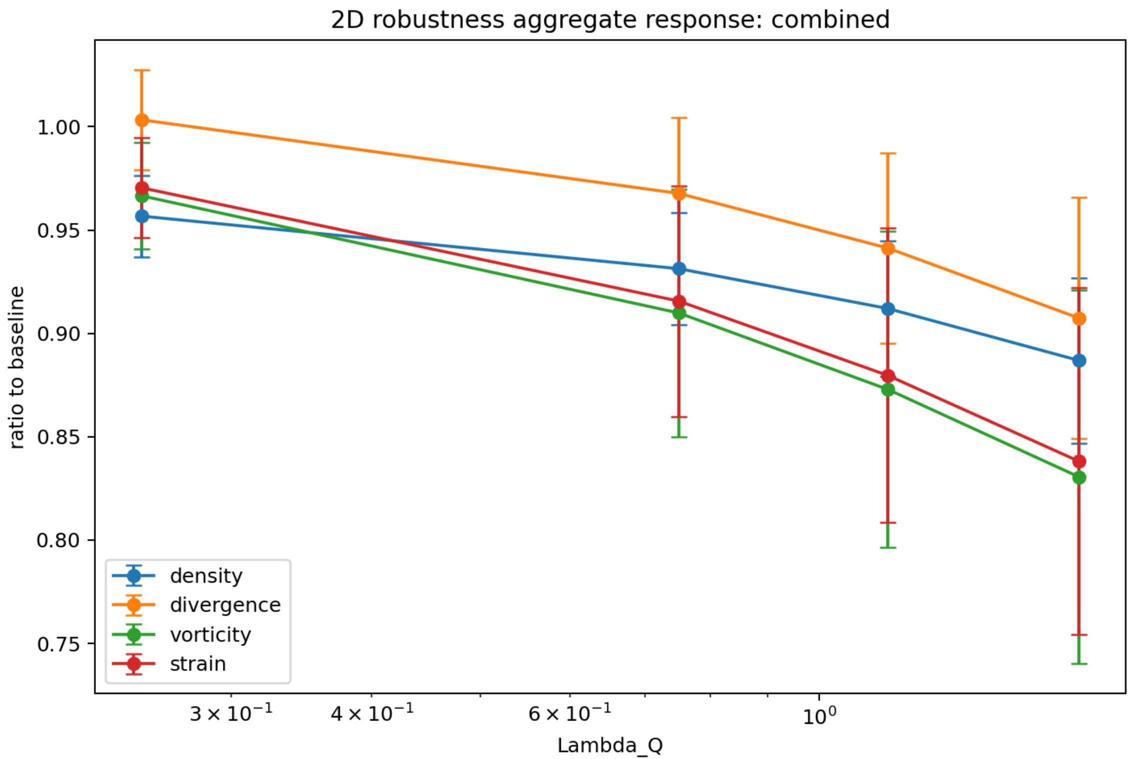

Figure 6. Combined channel: compression and shear/vorticity diagnostics decrease with increasing Lambda_Q.

13. 2D High-k Vorticity Suppression

The 2D tensor and combined channels suppress vorticity high-k content as Lambda_Q increases. This is the opposite of the 1D scalar-memory high-k growth, and it clarifies the dimensional lesson: 1D collapses compression and shear into one direction; 2D separates scalar compression from tensor shear.

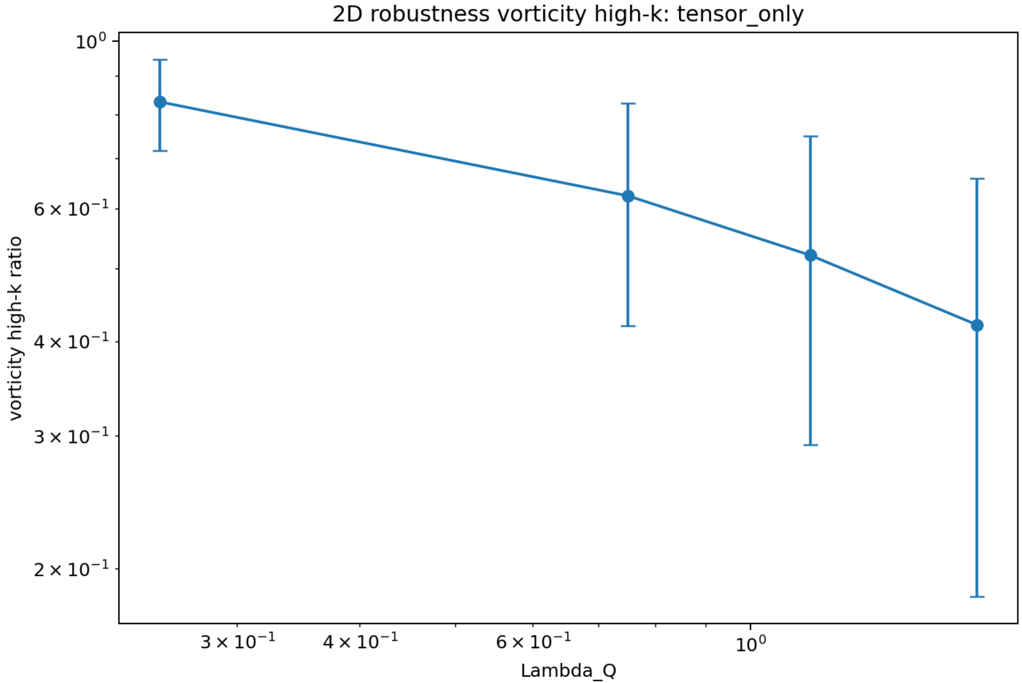

Figure 7. Tensor-only vorticity high-k response falls with Lambda_Q.

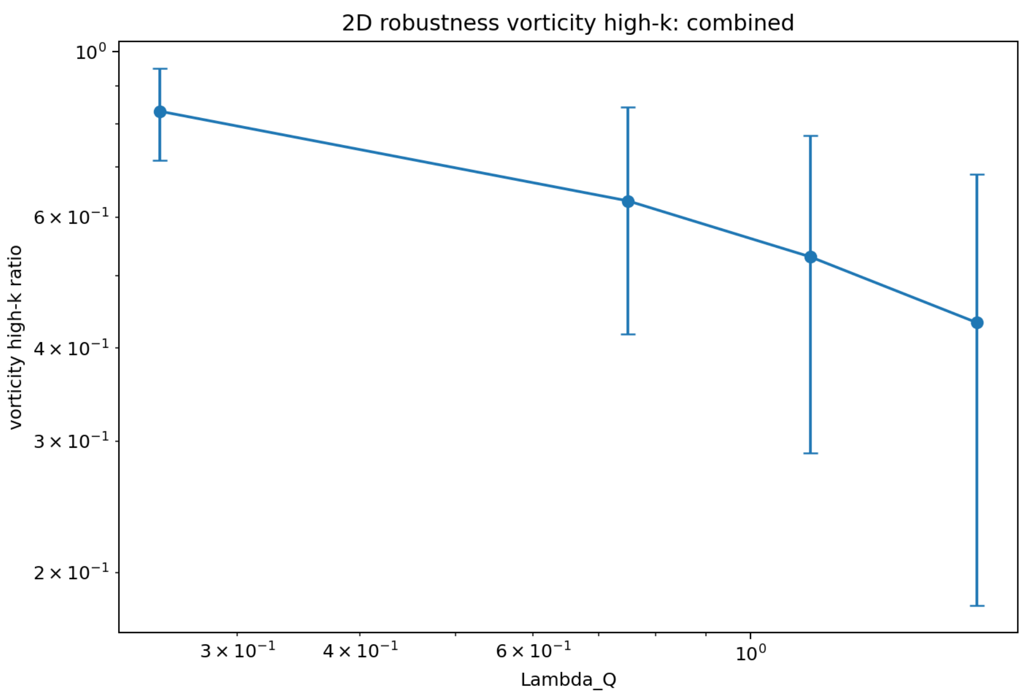

Figure 8. Combined channel vorticity high-k response also falls with Lambda_Q.

14. Correlation Summary

The strongest organizing correlations are the density and gradient response in 1D, and the vorticity/enstrophy/high-k response in the 2D tensor channel. Energy is deliberately not treated as a primary collapse metric because it remains more initial-condition sensitive.

Sector	Metric	mean corr	min	max
1D	rho_std_ratio	-0.915	-0.928	-0.899
1D	grad_ratio	0.901	0.883	0.925
1D	highk_ratio	0.780	-0.762	0.895
scalar_only	rho_std_ratio	-0.969	-0.974	-0.965
scalar_only	div_rms_ratio	0.874	0.572	0.965
scalar_only	vorticity_rms_ratio	0.909	-0.744	0.990
tensor_only	rho_std_ratio	-0.906	-0.923	-0.889
tensor_only	vorticity_rms_ratio	-0.939	-0.956	-0.928
tensor_only	enstrophy_ratio	-0.945	-0.964	-0.934
tensor_only	vort_high_k_ratio	-0.931	-0.960	-0.875
combined	rho_std_ratio	-0.905	-0.922	-0.885
combined	vorticity_rms_ratio	-0.938	-0.956	-0.928
combined	enstrophy_ratio	-0.945	-0.964	-0.934
combined	vort_high_k_ratio	-0.925	-0.962	-0.874

Read these signs structurally: in 1D, Lambda is associated with density smoothing and gradient/high-k growth. In 2D, Lambda_chi organizes scalar compression, while Lambda_Q organizes tensor shear/vorticity damping.

15. Interpretation and Guardrails

Supported toy-level statement
SFT lag stress behaves as a memory-bearing effective stress correction, not ordinary viscosity.
1D scalar lag memory is organized by Lambda = epsilon alpha tau / D and survives resolution, IC, and viscosity variation.
2D lag stress separates into a scalar compressive channel and a tensor shear-memory channel.
The scalar channel acts pressure-like and does not directly generate vorticity.
The tensor channel damps vorticity, enstrophy, strain, and vorticity high-k power in the 2D hero run.
The scalar/tensor closure is the lowest-order linear-response structure allowed by symmetry for coarse-grained lag overlap.

What v6 does not yet establish
It does not prove SFT or replace Navier-Stokes.
It does not derive epsilon, alpha, tau, or D from microscopic Planck-scale lag geometry.
It does not yet include a full energy/temperature closure, baroclinic vorticity, or 3D turbulence.
It does not yet compare quantitatively against a standard viscoelastic model such as Maxwell or Oldroyd-B.
It should be read as an SFT-motivated effective-fluid stress closure, not as a complete first-principles derivation.

16. Remaining Work

The numerical validation phase is now sufficient for a mature seed note. The next work should shift from sweeping phenomenological coefficients to deriving them or comparing the effective structure against standard continuum models.

Priority	Task	Purpose
1	Derive epsilon, alpha, tau, D from coarse-grained lag overlap structure	Turn the toy closure into a theory sector.
2	Relate Q_chi dynamics to Maxwell/Oldroyd-B viscoelastic stress	Show what is new and what is equivalent to known continuum physics.
3	Add energy/temperature closure and baroclinic terms	Allow richer 2D vorticity generation and thermodynamic response.
4	Run 3D pilot after coefficient logic improves	Test actual turbulence and vortex stretching.
5	Identify an empirical or simulation benchmark	Move from internal toy validation to external comparison.

Bottom line. V11.7 v6 closes the robustness checklist for the toy numerical phase and adds the effective-theory derivation of the scalar/tensor closure. The primary open problem is now microscopic coefficient derivation and external benchmarking.

17. Repro Pack Manifest

The accompanying v6 repro pack includes the scripts, core CSV outputs, plots, and final PDF used for this update. File names are normalized into 1D and 2D subfolders.

Folder	Contents
scripts/	sft_v117_big_validation_runner.py; sft_v117_2d_robustness_runner.py
data/1d_viscosity_full/	runs.csv; baselines.csv; summary_by_ic_N_nu_lambda.csv; correlations_by_ic_N_nu.csv
plots/1d_viscosity_full/	aggregate_response_by_lambda.png; aggregate_highk_by_lambda.png; regime_counts.png
data/2d_hero/	runs.csv; baselines.csv; summary_by_group.csv; correlations.csv; aggregate tables; config.json
plots/2d_hero/	aggregate_response_*; aggregate_vort_highk_*; regime_counts.png
docs/	SFT_V11_7_Lag_Field_Fluid_Stress_Seed_v6.pdf

End of v6 final effective-theory update.